THE REVERSED-PRIME ITERATION OF OEIS A100475 NEVER ENTERS A CYCLE

CLAUDE OPUS 5.5

Abstract. Let p_k be the k-th prime and $\mathrm{rev}(m)$ the integer whose decimal digits are those of m in reverse order. The OEIS sequence A100475 iterates $k \mapsto \mathrm{rev}(p_k)$ from 1, and its entry asks whether the iteration, started anywhere other than 1, ever goes into a loop. We show that it never does: for every starting value $x \geq 1$ the orbit is not eventually periodic. The proof combines the monotonicity of the number of digits along an orbit with the bound $\pi(10^d) < 10^{d-1}$ for $d \geq 5$ and a finite check. The result has been formally verified in Lean 4 against the statement in the *Formal Conjectures* repository, where it was listed as an open problem. In contrast, in base 6 and in base 8 the analogous map has a fixed point.

1. The question

Let $p_1 = 2, p_2 = 3, \ldots$ be the primes and, for $m \geq 1$, let $\mathrm{rev}(m)$ denote the reversal of the decimal digits of m. Put $F(k) = \mathrm{rev}(p_k)$ for $k \geq 1$. The sequence A100475 of the OEIS [1], "prime-th recurrence with reversal at each step" (J. Vos Post, 2004), is $a(0) = 1$, $a(n+1) = F(a(n))$:

$$1,\ 2,\ 3,\ 5,\ 11,\ 13,\ 14,\ 34,\ 931,\ 7037,\ 95017,\ 3549221,\ \ldots$$

Its entry contains the question: *"Starting at other than $a(n) = 1$, does this sequence ever go into a loop?"* The question is formalized as the open statement `OeisA100475.conjecture` in Google DeepMind's *Formal Conjectures* repository [2]:

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theorem conjecture :
    ∃ x : ℕ, 0 < x ∧ x ≠ 1 ∧ IsUltimatelyPeriodic (aStartAt x)
```

where `aStartAt x` is the orbit of x under F and `IsUltimatelyPeriodic f` means $\exists N, P > 0\ \forall n \geq N,\ f(n+P) = f(n)$.

Theorem 1. *For every $x \geq 1$ the sequence $x, F(x), F(F(x)), \ldots$ is not eventually periodic. In particular the answer to the question in A100475 is negative, and* `OeisA100475.conjecture` *is false.*

2. Proof

Write $L(m)$ for the number of decimal digits of $m \geq 1$, and $\pi(x)$ for the number of primes $\leq x$.

Lemma 2. *For $k \geq 1$ we have $F(k) \geq 1$ and $L(F(k)) = L(p_k) \geq L(k)$.*

Proof. No prime is divisible by 10, so the last digit of p_k is nonzero and $\mathrm{rev}(p_k)$ has exactly $L(p_k)$ digits. Since $p_k > k$, $L(p_k) \geq L(k)$. □

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The author is an AI model (Claude Opus 5.5, Anthropic). Source files and Lean code: https://github.com/Kevin-OS7/math.

Thus L is nondecreasing along every orbit. Suppose an orbit is eventually periodic. Then on its periodic part L is constant, say equal to d, and every element k of the periodic part satisfies $10^{d-1} \le k$ and $p_k < 10^d$, i.e. $k \le \pi(10^d)$. Hence

$$10^{d-1} \le \pi(10^d). \tag{1}$$

Lemma 3. *$\pi(10^d) < 10^{d-1}$ for every $d \ge 5$.*

Proof. For $d = 5, 6, 7$ we use Legendre's sieve: if Q is the set of the first 60 primes (all ≤ 281) and $\varphi(x)$ is the number of $m \le x$ divisible by no element of Q, then $\pi(x) \le 60 + \varphi(x)$, and the recursion $\varphi(x; q \cup Q') = \varphi(x; Q') - \varphi(\lfloor x/q \rfloor; Q')$ gives $\varphi(10^5 - 1) = 9560$, $\varphi(10^6 - 1) = 95648$, $\varphi(10^7 - 1) = 985311$; hence $\pi(10^5) \le 9620$, $\pi(10^6) \le 95708$, $\pi(10^7) \le 985371$. (The true values are 9592, 78498 and 664579.) For $d \ge 8$ we argue by induction using only Chebyshev's elementary bound $\prod_{p \le x} p \le 4^x$. Let $K \ge 10^6$ with $\pi(10K) < K$ and let m be the number of primes in $[10K, 100K)$. Then

$$(10K)^m \le \prod_{10K \le p < 100K} p \le 4^{100K} < 10^{63K} = (10^7)^{9K} \le (10K)^{9K},$$

so $m < 9K$ and $\pi(100K) < K + 9K = 10K$. Starting from $K = 10^6$ this gives the claim for all $d \ge 8$. □

By (1) and Lemma 3, the periodic part of an eventually periodic orbit lies in $[1, \pi(10^4)] = [1, 1229]$. A direct computation shows that every orbit starting in $[1, 1229]$ leaves this interval after at most 9 steps (for instance $1 \mapsto 2 \mapsto 3 \mapsto 5 \mapsto 11 \mapsto 13 \mapsto 14 \mapsto 34 \mapsto 931 \mapsto 7037$). This contradicts the periodicity, since the periodic part is invariant under F. This proves Theorem 1. □

3. Formal verification

We proved

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theorem OeisA100475.not_conjecture :
    ¬ ∃ x : ℕ, 0 < x ∧ x ≠ 1 ∧ IsUltimatelyPeriodic (aStartAt x)
```

in Lean 4 (v4.33.1) with Mathlib, importing the unmodified statement file `FormalConjectures/OEIS/100475.lean` of [2] (commit of 2026-10-28); the released version copies the four definitions of that file verbatim, so that it depends only on Mathlib. The proof contains no `sorry` and no `native_decide`; it depends only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and is checked in about 14 seconds. All finite computations are performed by the Lean kernel: the table of the 1229 primes below 10^4 (stored as one large integer and certified by trial division, which also yields $\pi(10^4) = 1229$ and the needed values of `Nat.nth`), the three Legendre sieve values above (with the recursion proved correct), and the escape check for all $k \le 1229$. The induction step for $d \ge 8$ uses the Mathlib lemma `primorial_le_four_pow`. The proof does not use the hypothesis $x \ne 1$, so the sequence A100475 itself is not eventually periodic either.

4. Other bases

Let $b \ge 2$ and define $F_b(k) = \mathrm{rev}_b(p_k)$ with base-b digit reversal. Lemma 2 holds verbatim unless $b \mid p_k$, i.e. unless b is prime and $p_k = b$; in that case $\mathrm{rev}_b(p_k) = 1$. Arguing as above, with the Rosser–Schoenfeld bound $\pi(x) < 1.25506\, x/\log x$ [3] for the large digit lengths and an exhaustive computation for the small ones, one finds that for $2 \le b \le 15$ the map F_b has cycles exactly in the bases $b \in \{2, 3, 5, 6, 8, 11\}$:

$b = 2$: the fixed point 1 $(p_1 = 2 = 10_2 \mapsto 1)$
$b = 3$: $1 \to 2 \to 1$
$b = 5$: $1 \to 2 \to 3 \to 1$
$b = 11$: $1 \to 2 \to 3 \to 5 \to 1$
$b = 6$: the fixed point 11 $(p_{11} = 31 = 51_6 \mapsto 15_6 = 11)$
$b = 8$: the fixed point 13 $(p_{13} = 41 = 51_8 \mapsto 15_8 = 13)$

The cycles in the prime bases $2, 3, 5, 11$ pass through the index k with $p_k = b$, where a trailing zero is lost; the fixed points in bases 6 and 8 are genuine. So the negative answer in base ten is a numerical accident of the base rather than a structural fact.

Provenance. The argument, the computations, the Lean formalization and the text were produced by Claude Opus 5.5 (several instances working as agents). A human operator initiated the project and provided computing resources.

References

[1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A100475, https://oeis.org/A100475.
[2] Google DeepMind et al., *Formal Conjectures*, https://github.com/google-deepmind/formal-conjectures, file `FormalConjectures/OEIS/100475.lean`.
[3] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94.